

5 Tips to Engineer Your Career

A guide for electronics engineering students who are just starting out their academic studies and also those who want to set themselves apart in a competitive career environment



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Last month, the idea about this article came to my mind just after watching the movie *3 Idiots*. By now, I guess you haven't been able to escape from being influenced either. The movie—a fairly serious take—essentially depicts how eager we are to embrace mindlessness and how keen we are to promote a culture where education is purely examination-centric without real-life application.

It's time that you get out of this 'aka idito' vagueness of just following the herd and passing the exam. Here is your 'why' and 'how to' guide.

Why focus on practical knowledge?

"There is a gap between engineering course content and the requirements of the engineering services industry," says Krishna Kant, senior programme manager, academic alliance programme-South Asia & Russia, EMC, from his vast talent nurturing experience.

C.P. Ravikumar, director-university relation, Texas Instruments, adds, "We perceive this gap during the process of interviews for regular as well

as internship positions." While our education system imparts knowledge on various technical/non-technical areas, it often falls short of meeting the expectations of the real world. "The biggest gap is the practical application and knowledge of industry practices related to the requirements of a particular role," says Lokesh Mehra, regional manager-corporate responsibility, Cisco South Asia.

One reason for this gap is a fundamental lacuna in the engineering education framework. This is the only profession where an individual goes from an academic programme directly into a job, with no prior on-the-job training. All the other fields of study require individuals to have a six-month to one-year long apprenticeship or internship before they start doing the real job.

"Getting fresh graduates ramped up quickly to productivity is a key concern across the industry ecosystem as new graduates sometimes take six months to a year to become productive. Ideally, we should overhaul engineering education and make its duration five years by including a mandatory six-months to one-year long apprenticeship," opines Jaswinder Ahuja, corporate vice president and managing director, Cadence Design Systems.

Vinay Shetty, country head-company business, ASUS (India), explains, "When fresh graduates enter the working environment, they have lots of theoretical knowledge which is

redundant, rather than the necessary practical knowledge."

Neelam Kumar, executive director, Aplab, points out, "Most of our institutions and universities lag behind the rest of the world in research output."

What should you do?

Here I take you through five boosters for your career.

Tip #1: Pay attention to basics

It may not sound important, but most of the interviewers will start with active and passive components, building blocks of electronics systems and nonetheless fault-finding techniques. "A strong foundation of the basics of electronics is must," says Raminder Singh Soin, managing director, Quad.

Strong fundamentals of electronics will help you to understand complex topics that you may get exposed to after joining the industry.

An electronics engineer is expected to have an excellent knowledge of electronic devices and RF, analogue, digital and especially CMOS design. This includes electrical fundamentals like signal integrity and power integrity. "Specifically, expertise in VLSI, VHDL, FPGA and ASIC design, signal processing, control systems, industrial manufacturing systems, power transmission, simulation and verification techniques is required," says Neeraj Varma, country manager-sales, Xilinx, India, Australia and New Zealand.

In terms of languages, one must be familiar with HDL (Verilog or VHDL), C and C++. Other skills that an electronics engineer must seek to possess are domain knowledge of microprocessors, control systems, embedded systems, and circuit and device testing.